Bump'em perturbation system: mains power requirements

IBMS lab, Hochschule Offenburg · for the building electrician · 21 October 2026 · prepared by Lorenzo Lucchini

1. The system

Four identical towers (A, B, C, D) stand around the GRAIL treadmill. Each tower pulls a rope attached to a person on the treadmill with an electric motor. The mains loads are only the power supplies listed below; all other parts run on their DC outputs (48 V motor driver, 24 V force amplifier). One control PC (laptop, own power supply) connects to the towers by USB.

Today: each tower's supplies are plugged into a power strip; there is **no wall socket per tower**, and we do not know which circuits and breakers the strips end up on. Nobody on our side works on the 230 V side.

2. Mains loads per tower (manufacturer data)

	Mean Well RSP-2000-48 (48 V motor supply)	**Mean Well RS-15-24** (24 V amplifier supply)
Rated output	48 V, 42 A, 2016 W	24 V, 0.625 A, 15 W
Input voltage / frequency	90–264 VAC (full power from 200 VAC), 47–63 Hz	85–264 VAC, 47–63 Hz
Input current, nameplate	**12.3 A** at 200–240 VAC	0.35 A at 100–240 VAC
Input current, typ. at full load	10 A at 230 VAC	0.25 A at 230 VAC
Inrush current (cold start, typ.)	**50 A**	**65 A** at 230 VAC
Earth leakage current	< 2 mA at 240 VAC	< 2 mA at 240 VAC
Power factor / efficiency	0.97 at 230 VAC full load / 92 %	– / 82 %
Mains connection	screw terminals L, N, PE (protection class I)	screw terminals L, N, PE; **open, not covered**
Other	internal fans; overload protection 105–125 %	no-load power < 0.5 W

Per tower, nameplate: $12.3 + 0.35 \approx$ **12.7 A** at 230 V. **All four towers, nameplate: ≈ 51 A.**
Sources: Mean Well datasheets RSP-2000 and RS-15, and the device labels.

3. Load actually drawn by this system

The nameplate is what the supplies *could* deliver. In this system the motor current is limited by the motor drivers' configuration to 9.9 A per motor, which caps what each tower can take:

Situation	per tower	4 towers together
Switched on, idle (no pull; not specified by Mean Well)	small	small
Holding the rope slightly taut (normal use)	< 50 W	< 200 W
Maximum pull, current setup (motor 9.9 A at 48 V)	≤ 520 W (≈ 2.3 A)	$\leq$ **2.1 kW (≈ 9.3 A)**
Possible future setting (software force limit 200 N, ≈ 16.8 A motor current)	≤ 870 W (≈ 3.8 A)	≤ 3.5 kW (≈ 15 A)

Maximum pulls are short: a pull lasts at most 0.6 s, typically a few per minute. Energy from braking the motor is burnt in a resistor on the 48 V side (brake chopper); nothing is fed back into the mains. Estimate by us from the drive settings, not a measurement; we can measure with a plug-in power meter if useful.

4. What we would like, and our questions

1. **A supply point at each tower:** one earthed 230 V socket (Schuko, 16 A) or equivalent per tower, so each tower has its own connection instead of chained power strips.

   Question: Which circuits / breakers can feed the four towers? Our suggestion: at least two circuits (two towers each), so the possible future 3.5 kW and the inrush below are covered with reserve.

2. **Inrush:** each tower draws up to about 50 A (RSP-2000) plus 65 A (RS-15) for a few milliseconds at switch-on. We plan to switch the towers on one at a time.

   Question: Is the breaker characteristic (B or C) suitable for this inrush?

3. **Residual current devices:** leakage up to 2 mA per supply, i.e. up to 4 mA per tower and 16 mA for all four.

   Question: How many towers per 30 mA RCD, and which RCD type for switch-mode supplies with power-factor correction?

4. **Protective earth:** each supply has a PE terminal. On tower B the PE connection to the brake chopper's earth terminal (terminal 3) is still to be made, as on the other towers.

   Question: Please check PE on all four towers.

5. **Touch protection:** the four RS-15-24 have open 230 V screw terminals.

   Question: Please fit a cover (or a suitable enclosure) on each.

6. **Emergency stop:** our planned emergency-stop button switches off the 48 V outputs through the supplies' remote on/off input. It does *not* disconnect the mains.

   Question: Is a mains emergency-off needed for this setup in the lab? Is an electrical safety inspection (DGUV V3) needed before use with test persons?

Tower layout: A left, B front, C right, D back of the treadmill. Datasheets and the full technical documentation can be provided on request.